Examination Management System

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Id no: 444-5024
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AGENDA

- Introduction & Objective
- SRS & System Architecture
- Gantt Chart
- Data Dictionary
- DFD
- ER-DIGRAM
- Interface
- Expected Report
- References or Bibliography
- Future Scope

INTRODUCTION

- Online Examination System is a web-based application
- It digitizes the traditional examination process
- It reduces manual work, time consumption, and human errors
- The system is secure, efficient, and user-friendly

OBJECTIVES OF THE PROJECT

- To conduct examinations online
- To provide secure authentication using JWT
- To generate results automatically
- To maintain separate dashboards for students and teachers
- To Manage Students, analyze performance and collect feedback

SYSTEM ARCHITECTURE

Frontend:

- HTML, CSS, JavaScript

Backend:

- Node.js, Express.js
- RESTful APIs

Database:

- MongoDB

SRS – HARDWARE REQUIREMENTS

- Computer or Laptop
- Minimum 4 GB RAM
- Stable Internet Connection

SRS – SOFTWARE REQUIREMENTS

- **Operating System:** Windows / Linux / macOS
- **Web Browser:** Google Chrome / Microsoft Edge
- **Runtime Environment:** Node.js
- **Database:** MongoDB
- **Code Editor:** Visual Studio Code

PROCESS LOGIC

Student:

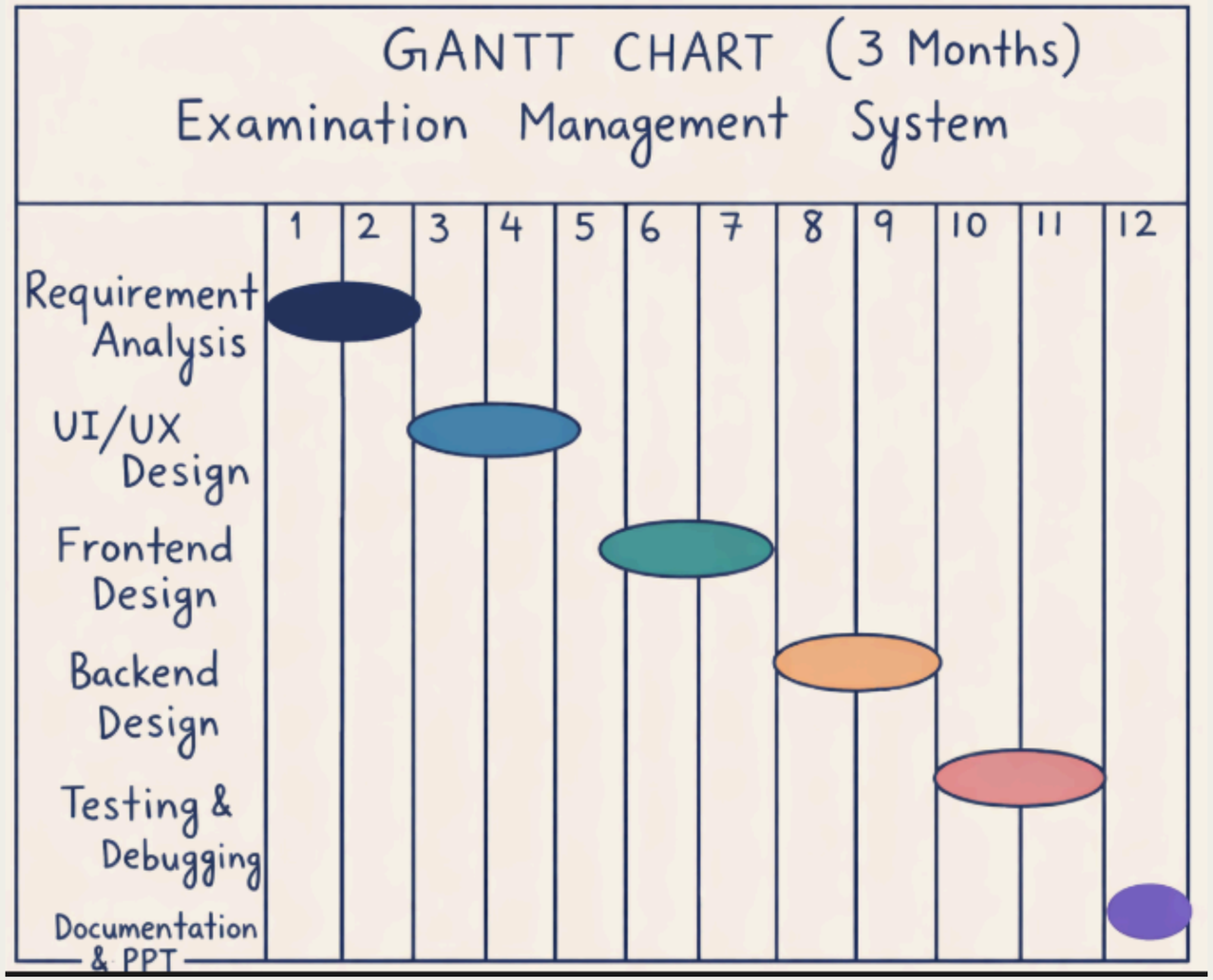
- Student logs into the system
- Authentication is verified
- Student is redirected to dashboard
- Then he/she open exam publish link on another tab which is provided by their teacher.
- Then the exam will appear on their screen.
- Then they will start the exam and Questions are displayed to their screen.
- After Attempting Question they will submit their exam
- Result will evaluated automatically
- They will see their results and give the feedback.

PROCESS LOGIC

Teacher:

- Teacher logs into the system
- Authentication is verified
- Teacher is redirected to dashboard
- Then they can create and Publish Exam, View Statistics, Delete Exam, Manage students, Manage feedbacks

GANNT CHART


GANTT CHART (3 Months)
Examination Management System
1
2
3
4
5
6
7
8
9
10
11
12
Requirement Analysis
UI/UX Design
Frontend Design
Backend Design
Testing & Debugging
Documentation & PPT

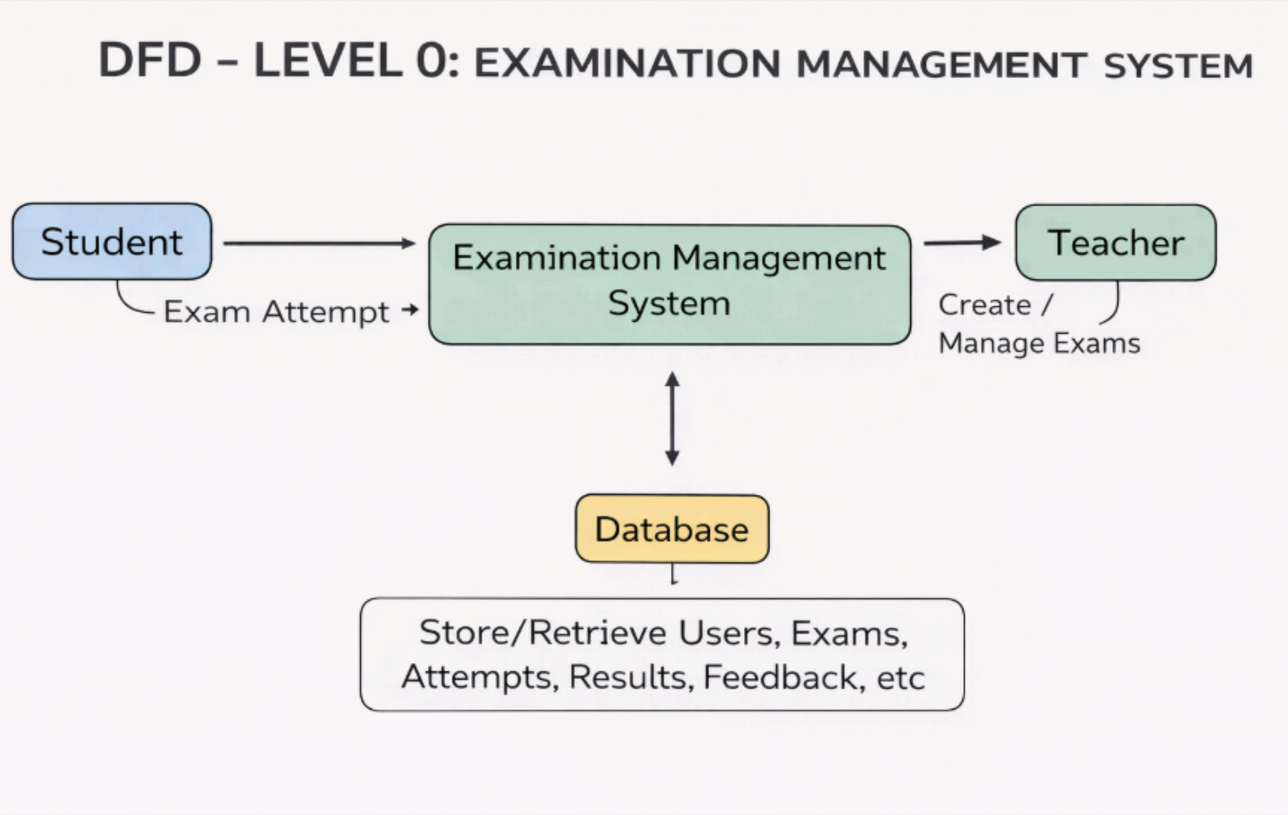
DATA DICTIONARY

Table Name	Field Name	Data Type	Description
USERS	_id (PK)	ObjectId	System-generated unique identifier for each user
USERS	first_name	String	First name of the user
USERS	last_name	String	Last name of the user
USERS	email	String	Unique email address used for login
USERS	password	String	Encrypted / hashed password of the user
USERS	user_type	String	Role of the user (Student / Teacher)
USERS	date_of_birth	Date	Date of birth of the user
USERS	gender	String	Gender of the user
USERS	institution	String	Name of the institution
USERS	field_of_study	String	Academic field or department
USERS	createdAt	Date	System-generated record creation timestamp
USERS	updatedAt	Date	System-generated record last update timestamp
EXAMS	_id (PK)	ObjectId	System-generated unique identifier for each exam
EXAMS	exam_id	String	Human-readable unique exam code
EXAMS	title	String	Title of the examination
EXAMS	description	String	Detailed description of the exam
EXAMS	target_audience	String	Intended group of students
EXAMS	scheduled_at	Date	Date and time when exam is scheduled
EXAMS	duration_minutes	Number	Total duration of the exam in minutes
EXAMS	status	String	Current exam status (Draft / Published)
EXAMS	created_by (FK)	ObjectId	Teacher (user) who created the exam
EXAMS	createdAt	Date	System-generated exam creation time
EXAMS	updatedAt	Date	System-generated exam update time

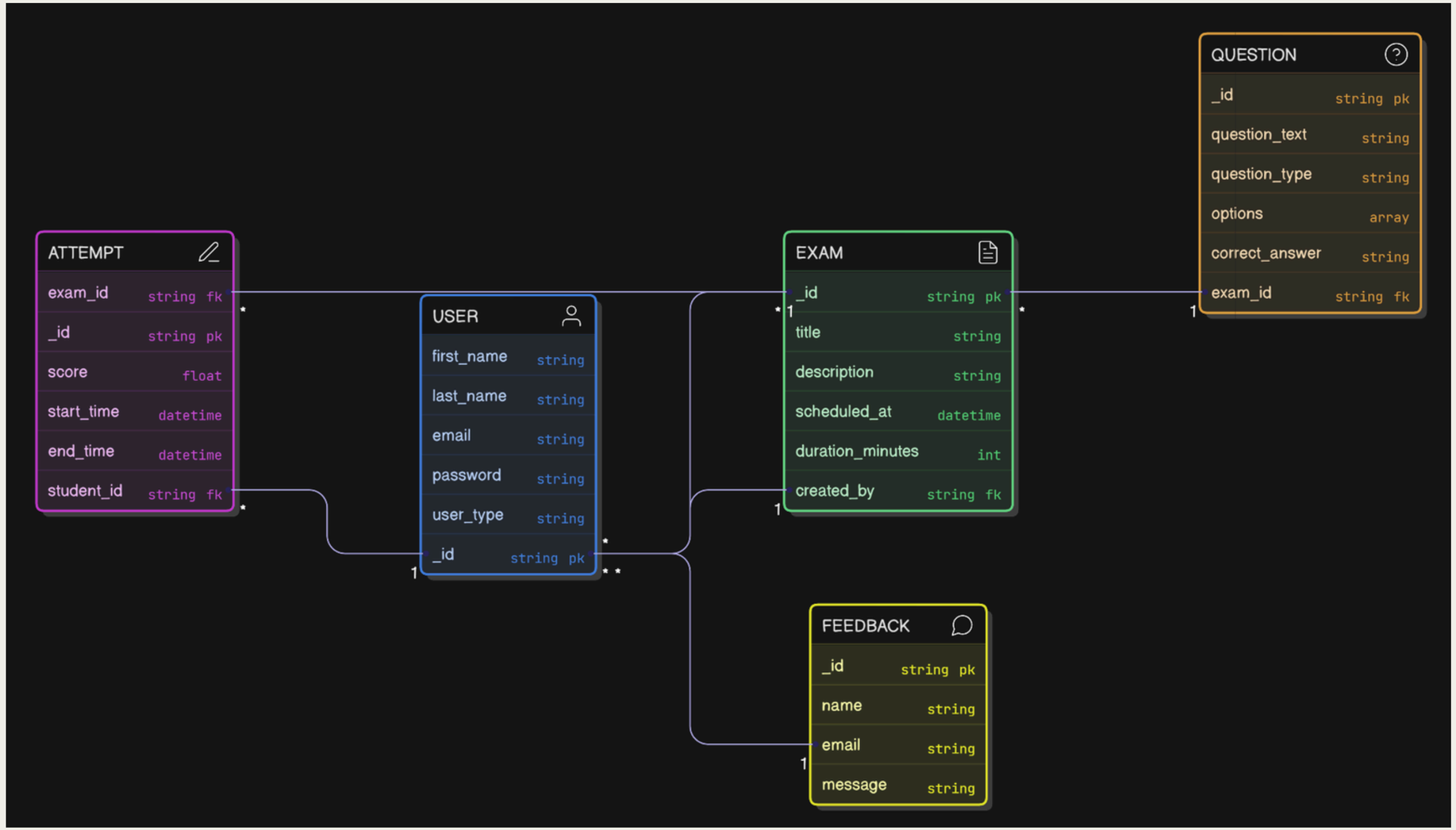
DATA DICTIONARY

Table Name	Field Name	Field Name	Description
QUESTIONS	_id (PK)	ObjectId	System-generated unique identifier for each question
QUESTIONS	exam_id (FK)	String	Exam ID to which the question belongs
QUESTIONS	question_text	String	Text/content of the question
QUESTIONS	question_type	String	Type of question (MCQ / Direct)
QUESTIONS	points	Number	Marks assigned to the question
QUESTIONS	options	Array	List of options for MCQ type questions
QUESTIONS	correct_answer	String	Correct answer for direct questions
QUESTIONS	correct_options	Array	Correct option(s) for MCQ questions
QUESTIONS	attachment	String	Media attachment (audio/video) for future use
QUESTIONS	createdAt	Date	System-generated question creation time
QUESTIONS	updatedAt	Date	System-generated question update time
ATTEMPTS	_id (PK)	ObjectId	System-generated unique identifier for each attempt
ATTEMPTS	exam_id (FK)	String	Exam ID attempted by the student
ATTEMPTS	student_id (FK)	ObjectId	Student who attempted the exam
ATTEMPTS	start_time	Date	Time when the exam attempt started
ATTEMPTS	end_time	Date	Time when the exam attempt ended
ATTEMPTS	score	Number	Total marks obtained by the student
ATTEMPTS	answers	Array	List of answers submitted by the student
ATTEMPTS	ip_address	String	IP address of the student during exam
ATTEMPTS	coordinates	GeoPoint	Geographical location of the student
ATTEMPTS	completed	Boolean	Indicates whether exam attempt is completed
ATTEMPTS	createdAt	Date	System-generated attempt creation time
ATTEMPTS	updatedAt	Date	System-generated attempt update time
FEEDBACK	_id (PK)	ObjectId	System-generated unique identifier for feedback
FEEDBACK	name	String	Name of the user submitting feedback
FEEDBACK	email	String	Email address of the feedback provider
FEEDBACK	message	String	Feedback message or suggestion
FEEDBACK	createdAt	Date	System-generated feedback submission time
FEEDBACK	updatedAt	Date	System-generated feedback update time

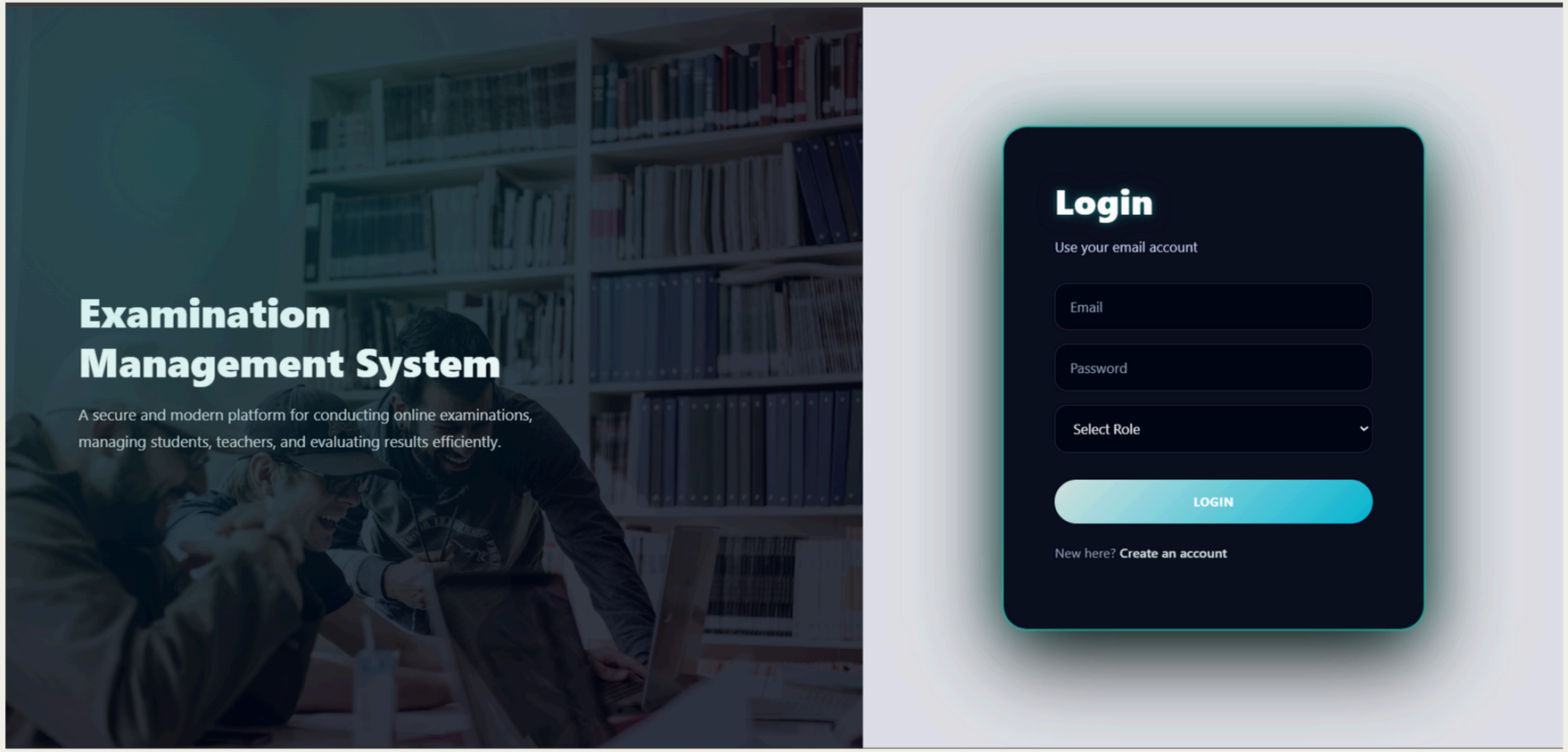
DFD


DFD – LEVEL 0: EXAMINATION MANAGEMENT SYSTEM
Student
Exam Attempt
Examination Management System
Teacher
Create / Manage Exams
Database
Store/Retrieve Users, Exams, Attempts, Results, Feedback, etc

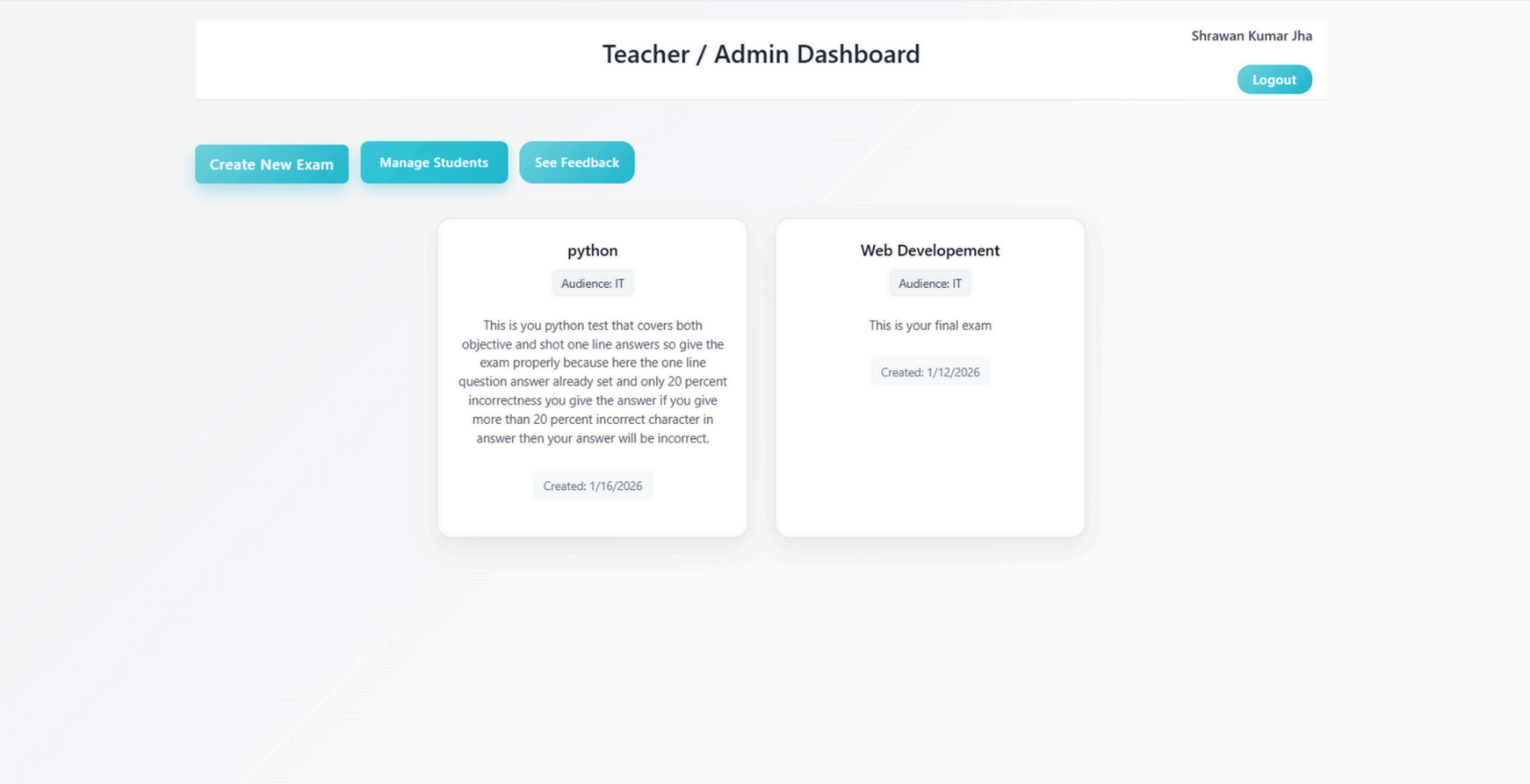
ER-DIGRAM


ATTEMPT
exam_id string fk
_id string pk
score float
start_time datetime
end_time datetime
student_id string fk
USER
first_name string
last_name string
email string
password string
user_type string
_id string pk
EXAM
_id string pk
title string
description string
scheduled_at datetime
duration_minutes int
created_by string fk
QUESTION
_id string pk
question_text string
question_type string
options array
correct_answer string
exam_id string fk
FEEDBACK
_id string pk
name string
email string
message string

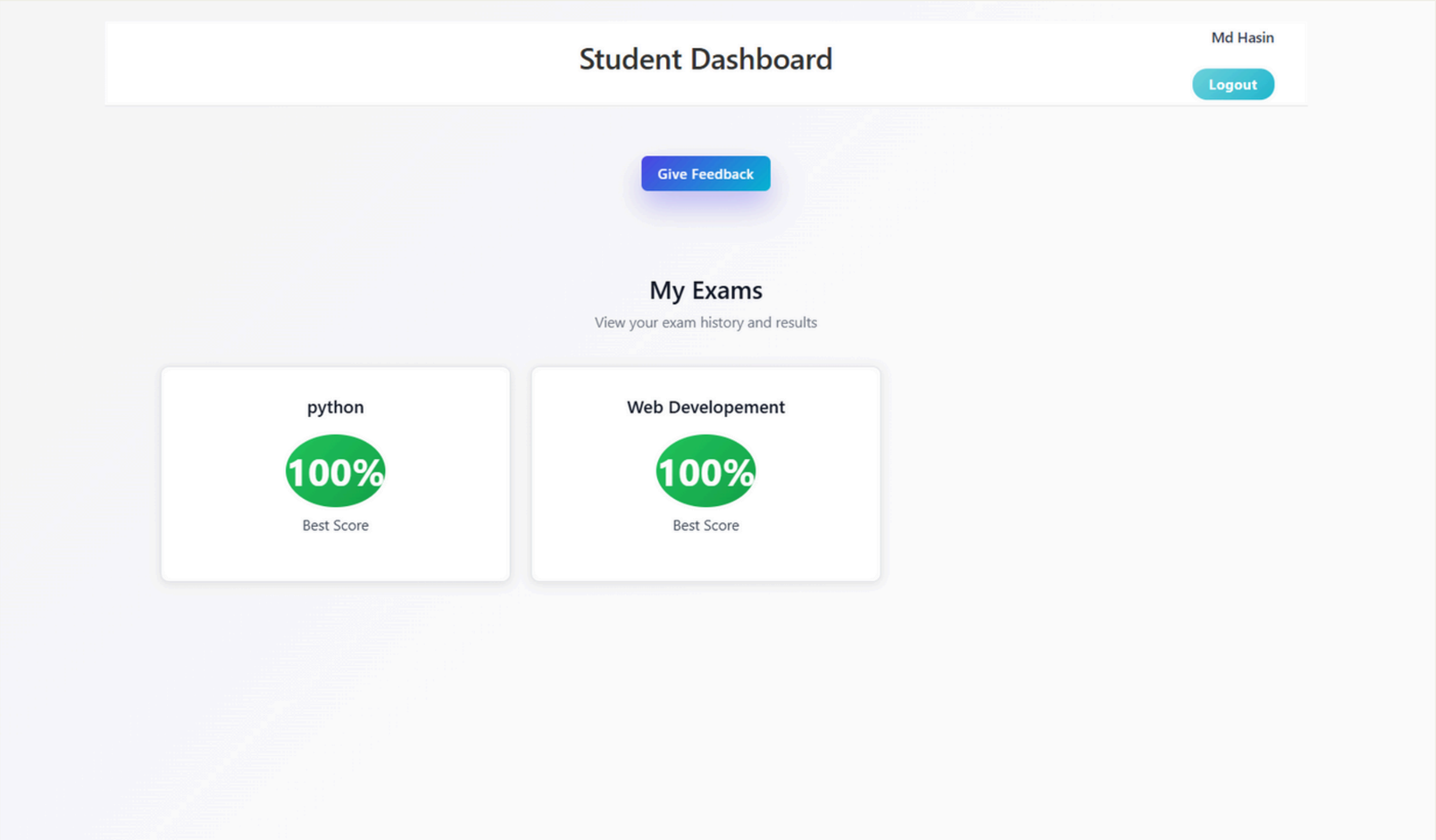
INTERFACE


Examination
Management System
A secure and modern platform for conducting online examinations, managing students, teachers, and evaluating results efficiently.
Login
Use your email account
Email
Password
Select Role
LOGIN
New here? Create an account

INTERFACE


Teacher / Admin Dashboard
Shrawan Kumar Jha
Logout
Create New Exam
Manage Students
See Feedback
python
Audience: IT
This is you python test that covers both objective and shot one line answers so give the exam properly because here the one line question answer already set and only 20 percent incorrectness you give the answer if you give more than 20 percent incorrect character in answer then your answer will be incorrect.
Created: 1/16/2026
Web Developement
Audience: IT
This is your final exam
Created: 1/12/2026

INTERFACE


Student Dashboard
Md Hasin
Logout
Give Feedback
My Exams
View your exam history and results
python
100%
Best Score
Web Developement
100%
Best Score

EXPECTED REPORT

- Student Result Report
- Exam-wise Performance Report
- Student Attempt History
- Feedback Analysis Report

REFERENCES

- Node.js Official Documentation
- MongoDB Official Documentation
- Express.js Documentation
- MDN Web Docs
- OpenAI – ChatGPT

FUTURE SCOPE

- AI-based cheating detection
- Live proctoring system
- Mobile Application
- Certificate Generation

Thank You

For your attention

Shoka